

Compatible-Frame Natural-Tile Geometry Specification

Stage 11B: Source-Bound Gauge Replay, Dynamic Tile and Window Descriptors, and Explicit Frame Compatibility

mdstats

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Purpose and stage boundary

Stage 11B maps one already certified periodic natural tiling onto compatible trajectory or ensemble frames without changing its scientific identity:

```
certified natural tiling + exact reference geometry
+ atomistic frame collection
+ source-bound atomic connectivity
+ projected framework topology catalog
-> replay canonical periodic vertex gauge
-> map fixed scientific faces, windows, and tiles
-> compute frame-dependent geometry
-> preserve explicit unresolved frame states
```

Runtime/API target:

mdstats 0.19.29a0

Primary module:

mdstats/analysis/tiling_geometry_frames.py

Stage 11B is a descriptive geometry layer. It does **not** rediscover rings, faces, tiles, or natural-tiling identity in each frame. A frame is mapped only when its exact projected framework graph agrees with the source graph of the certified `PeriodicCellComplex`.

Scientific invariants

The implementation preserves the following separation.

Persistent scientific identity

The following remain fixed across all mapped frames:

- primitive-ring keys;
- scientific face placements and orientations;
- tile attaching maps;
- topological-window identity;
- translation-labelled tile adjacency; and
- tile labels and orbit identity.

Frame geometry is evidence attached to those identities. It cannot add, remove, or relabel scientific cells.

Frame compatibility

Compatibility is stronger than matching atom counts. For each selected frame:

1. its atomic connectivity state is projected using the same `FrameworkMapping` and projection policy as the supplied `TopologyCatalog`;
2. the resulting `FrameworkTopology.graph_digest` must equal `PeriodicCellComplex.topology_graph_digest`;
3. the atomic periodic image gauge must be reconstructible from the declared unique-minimum-image connectivity convention; and
4. the reconstructed tile surfaces must remain nondegenerate, orientation preserving, and volume closing.

Failure at one frame does not invalidate other frames. It produces an explicit per-frame status.

Nonplanar dynamic faces

Reference scientific faces are exact planar polygons. Thermally distorted frame faces need not be planar. Stage 11B retains their fixed boundary order and uses a deterministic boundary-center fan surface for area, volume, and first-moment integration. Planarity and projected-aperture fields are descriptive diagnostics; they do not redefine the face.

External methods and original construction

The translation-labelled periodic quotient convention is inherited from Chung, Hahn, and Klee [1]. Natural-tile identity is inherited from the Stage-10/11 construction based on Blatov and coauthors [2]. Singular-value decomposition for best-fit plane diagnostics is standard numerical linear algebra background and is not treated as a borrowed domain algorithm.

The following mechanisms are project-specific `mdstats` constructions:

- exact replay of the composed atomic, relevant-subgraph, and projected-framework image gauges used by the existing connectivity/topology pipeline;
- trajectory anchor alignment to the collection's unwrapped fractional coordinate without changing intrinsic geometry;
- independent ensemble wrapping with no cross-frame continuity assumption;
- fixed scientific side reconstruction from source-bound ring walks and attaching translations;
- deterministic nonplanar boundary-center fan integration;
- explicit separation of topology mismatch, connectivity/geometry mismatch, and degenerate mapped geometry; and
- source-replayed persistent frame catalogs with independent geometry and connectivity binding digests.

No external trajectory pore-tracking, deforming-cell tessellation, dynamic Voronoi, or maximum-clearance algorithm is claimed.

Public API

```
map_tiling_geometry_to_frames(  
    reference_geometry: TilingGeometryCatalog,  
    complex_: PeriodicCellComplex,  
    embedding: PeriodicNetEmbedding,  
    ring_index: PrimitiveRingIndex,
```

```

collection: AtomisticFrameCollection,
connectivity: AtomicConnectivityResult,
topology_catalog: TopologyCatalog,
*,
frame_indices: Sequence[int] | None = None,
options: FrameTilingGeometryOptions | None = None,
resources: FrameTilingGeometryResources | None = None,
) -> FrameTilingGeometryCatalog

```

Required source equalities include

```

reference_geometry.periodic_cell_complex_digest == complex_.digest
reference_geometry.periodic_net_embedding_digest == embedding.digest
reference_geometry.primitive_ring_catalog_digest == ring_index.catalog_digest
complex_.topology_graph_digest == ring_index.topology_graph_digest
collection.frame_semantics == topology_catalog.frame_semantics

```

The connectivity result must declare

```
metadata["unique_minimum_image_only"] == True
```

because Stage 11B replays the physical atomic image gauge under that exact convention. The implementation does not guess a projected-net nearest image.

Periodic gauge replay

Let atom i have wrapped fractional coordinate $\bar{\mathbf{f}}_i$ in one frame. The atomic connectivity state stores canonical edge shifts $\mathbf{s}_{ij}$. The physical minimum-image atomic edge has raw shift $\mathbf{m}_{ij}$. The atomic canonicalization gauge $\mathbf{a}_i$ satisfies

$$\mathbf{s}_{ij} = \mathbf{m}_{ij} + \mathbf{a}_j - \mathbf{a}_i.$$

The framework projection then applies two additional integer gauges already encoded by its deterministic normalization:

- $\mathbf{h}_i$: relevant-subgraph gauge;
- $\mathbf{g}_i$: final projected-framework gauge.

The frame coordinate of a canonical framework vertex is reconstructed as

$$\mathbf{q}_i = \bar{\mathbf{f}}_i + \mathbf{a}_i + \mathbf{h}_i + \mathbf{g}_i + \mathbf{G},$$

where $\mathbf{G} \in \mathbb{Z}^3$ is one global placement shift.

For a trajectory, $\mathbf{G}$ aligns one deterministic anchor vertex with the collection's unwrapped fractional coordinate. This preserves whole-cell continuity. For an ensemble, $\mathbf{G}$ independently places the anchor in the reference cell; no temporal continuity is inferred.

A lifted vertex reference $(i, \mathbf{n})$ is then mapped to

$$\mathbf{f}_{i,\mathbf{n}} = \mathbf{q}_i + \mathbf{n}, \quad \mathbf{x}_{i,\mathbf{n}} = \mathbf{f}_{i,\mathbf{n}}H + \mathbf{o},$$

where rows of H are the instantaneous lattice vectors and $\mathbf{o}$ is the frame origin.

Fixed scientific face mapping

For every reference `TileFaceGeometry`, Stage 11B reconstructs the source ring walk, face placement shift, tile-side attaching shift, and orientation. The resulting ordered lifted references must replay the exact reference polygon before frame coordinates are evaluated.

For Cartesian face vertices $\mathbf{x}_k$, define their arithmetic center

$$\mathbf{c}_f = \frac{1}{n} \sum_{k=0}^{n-1} \mathbf{x}_k.$$

The deterministic fan consists of triangles

$$(\mathbf{c}_f, \mathbf{x}_k, \mathbf{x}_{k+1}).$$

The stored face area is

$$A_f = \frac{1}{2} \sum_k \| (\mathbf{x}_k - \mathbf{c}_f) \times (\mathbf{x}_{k+1} - \mathbf{c}_f) \|.$$

Its area-weighted oriented vector determines the reported unit normal. Every fan triangle must exceed the declared degeneracy tolerance.

Planarity diagnostics

A best-fit plane is obtained from the least singular vector of the centered vertex matrix. Stage 11B records

$$\delta_{\text{rms}} = \sqrt{\frac{1}{n} \sum_k d_k^2}, \quad \delta_{\text{max}} = \max_k |d_k|,$$

where d_k is signed distance to the best-fit plane.

Vertices are projected into that plane. The reported aperture witness is the minimum in-plane distance from the projected polygon center to its boundary segments. `planar_aperture_certified` is true only when the projected polygon is strictly convex, contains the witness, and δ_{max} does not exceed the configured planarity tolerance.

This flag certifies only the stored planar witness. It is not a maximum flexible portal radius.

Dynamic tile geometry

A mapped tile retains the source tile's ordered side indices. Unique lifted vertex identities are gathered from all its sides.

A Cartesian reference point $\mathbf{r}$ is chosen as the arithmetic mean of the unique tile vertices. Every oriented face-fan triangle $(\mathbf{c}_f, \mathbf{x}_k, \mathbf{x}_{k+1})$ defines one signed tetrahedron:

$$V_k = \frac{1}{6} (\mathbf{c}_f - \mathbf{r}) \cdot [(\mathbf{x}_k - \mathbf{r}) \times (\mathbf{x}_{k+1} - \mathbf{r})].$$

The signed tile volume and volume centroid are

$$V = \sum_k V_k, \quad \mathbf{c}_V = \frac{1}{V} \sum_k V_k \frac{\mathbf{r} + \mathbf{c}_f + \mathbf{x}_k + \mathbf{x}_{k+1}}{4}.$$

A mapped tile is accepted only when orientation is preserved and $V > 0$ beyond tolerance. Derived descriptors are:

- absolute volume;
- surface area;
- equivalent-sphere radius;
- Wadell sphericity;
- maximum pairwise lifted-vertex distance; and
- Cartesian and fractional volume centroids.

The sum of all tile volumes must close to the instantaneous primitive-cell volume within the configured absolute plus relative tolerance.

Dynamic windows and adjacency

The scientific `TopologicalWindow` catalog remains unchanged. For each window, the two mapped tile sides must agree geometrically after applying the stored translation relation. Stage 11B reports:

- Cartesian window center;
- mean side area;
- side-area mismatch;
- minimum of the two projected aperture witnesses;
- maximum planarity RMS and maximum deviation; and

- a planar-aperture certificate only when both sides certify it.

Translation-labelled adjacency arcs are inherited unchanged from the reference geometry. Stage 11B does not infer portal accessibility or crossing events.

Persistent result model

Per-frame status

```
FrameTilingGeometryStatus(
    MAPPED,
    TOPOLOGY_MISMATCH,
    CONNECTIVITY_GEOMETRY_MISMATCH,
    DEGENERATE_GEOMETRY,
)
```

- MAPPED: complete face, window, and tile geometry with volume closure.
- TOPOLOGY_MISMATCH: exact projected graph differs from the certified source.
- CONNECTIVITY_GEOMETRY_MISMATCH: graph identity matches, but the declared atomic image gauge cannot be reconciled with frame geometry.
- DEGENERATE_GEOMETRY: the gauge maps, but a face/tile/window is degenerate, orientation reversing, geometrically inconsistent, or fails volume closure.

Unmapped frames contain no partial geometry.

Main records

```
FrameTileFaceGeometry(...)
FrameNaturalTileGeometry(...)
FrameWindowGeometry(...)
MappedTilingFrame(...)
FrameTilingGeometryCatalog(...)
```

`FrameTilingGeometryCatalog.tile_metric(tile_index, metric)` returns a frame-aligned NumPy array and uses NaN for unresolved frames. Supported metrics are

```
volume
surface_area
equivalent_sphere_radius
sphericity
diameter
```

Options and resources

```
FrameTilingGeometryOptions(
    degeneracy_tolerance=1.0e-12,
    planarity_tolerance=1.0e-3,
    window_match_tolerance=1.0e-8,
    volume_closure_relative_tolerance=1.0e-8,
    volume_closure_absolute_tolerance=1.0e-8,
)

FrameTilingGeometryResources(
    max_frames=100_000,
    max_vertices=100_000,
    max_tile_faces=1_000_000,
    max_vertex_instances=10_000_000,
    max_pair_distance_tests=100_000_000,
)
```

Frame count, framework vertices, tile sides, and total mapped face-vertex work are preflighted before per-frame mapping begins. Tile diameter work is checked before its pairwise distance loop. Resource failure never returns a partial persistent catalog.

Serialization and binding

The persistent catalog binds independently to:

- reference geometry;
- scientific periodic cell complex;
- authoritative embedding;
- primitive-ring catalog;
- topology catalog;
- selected collection geometry; and
- selected atomic-connectivity states.

Collection binding includes frame semantics, selected frame positions and IDs, active atomic numbers, periodicity, cells, origins, and active fractional coordinates. Connectivity binding includes the exact definition, resolved scope, and selected state digests.

Loading is deterministic source replay:

```
FrameTilingGeometryCatalog.from_dict(  
    payload,  
    reference_geometry=reference_geometry,  
    complex_=complex_,  
    embedding=embedding,  
    ring_index=ring_index,  
    collection=collection,  
    connectivity=connectivity,  
    topology_catalog=topology_catalog,  
)
```

A payload is rejected when any stored field disagrees with the rebuilt result, even when an outer digest is recomputed.

Edge cases and failure policy

The first backend rejects or leaves explicit per-frame outcomes for:

- missing or non-unique selected frames;
- nonmonotonic trajectory frame selection;
- mismatched frame IDs or frame semantics;
- source digest disagreement;
- atom ordering incompatible with the source embedding;
- connectivity not using the unique-minimum-image convention;
- changed projected framework topology;
- a noninteger trajectory anchor reconciliation;
- inconsistent physical and canonical atomic edge images;
- degenerate boundary-center fan triangles;
- reversed or zero-volume tiles;
- inconsistent two-sided window geometry;
- failure of instantaneous cell-volume closure; and
- declared resource exhaustion.

The following remain outside Stage 11B:

- automatic atom correspondence between independently reordered structures;
- topology changes that require a new natural tiling;
- nonconvex or self-intersecting dynamic tile surfaces;
- explicit linker or van der Waals surfaces;
- chemistry-aware radius policies;
- maximum-clearance cage or portal searches;
- global free-volume connectivity;
- occupancy and portal-crossing event detection; and

- kinetic or energetic transport models.

Focused validation requirements

The Stage-11B focused suite must cover:

1. full mapping of the real-LTA ten-tile scientific complex;
2. exact isotropic scaling of volumes, areas, and lengths;
3. invariance to independent integer atom wrapping;
4. whole-cell trajectory anchor continuity;
5. descriptive retention of thermally nonplanar faces;
6. explicit topology-mismatch frames and NaN metric alignment;
7. independent ensemble wrapping;
8. transactional resource preflight;
9. source replay and tamper rejection; and
10. wrong-source and unsupported-metric rejection.

References

- [1] S. J. Chung, T. Hahn, and W. E. Klee, “N-dimensional crystallography,” *Acta Crystallographica A* **40**, 42–50 (1984), doi:10.1107/S010876738400010X.
- [2] V. A. Blatov, O. Delgado-Friedrichs, M. O’Keeffe, and D. M. Proserpio, “Three-periodic nets and tilings: natural tilings for nets,” *Acta Crystallographica A* **63**, 418–425 (2007), doi:10.1107/S0108767307038287.